

Q2.7

list1 = [1 → 2 → 4]

list2 = [1 → 3 → 4]

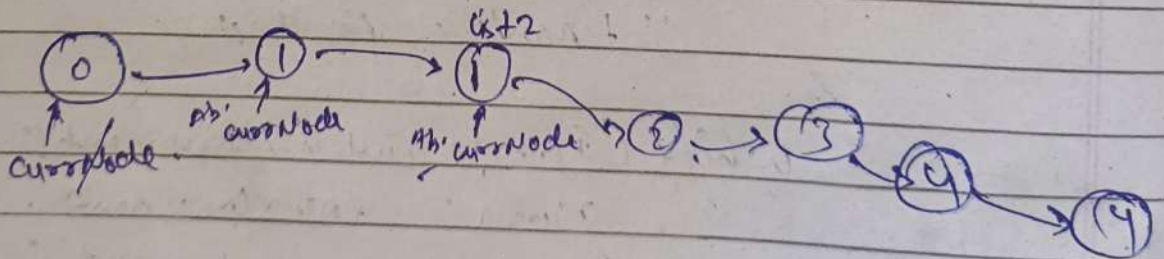
Sabse pehle base case :-

```
if (list1 == null) {
    return list2
}
```

```
if (list2 == null) {
    return list1
}
```

Ab hume two list ko merge करना hai.

Sabse pehle hum ek currNode create karenge.
jisme 0 store hoga.



Ab loop chalenge

```
while (list1 != null && list2 != null) {
    if (list1.data <= list2.data) {
        currNode.next = list1
        list1 = list1.next
    }
}
```

matlab agr list1 ka data small hai list2 ke data se. toh currNode ke next me list1 ko dal denge or list1 ko age bhejenge list1.next me.

To matlab ab $0 \rightarrow 1$ ko point kar raha hai.

ab else case.

Else (list2.data < list1.data)

currNode.next = list2.

list2 = list2.next.

→ Isatrak karte karte ek list me null ho jayega or hm loop se bahar ho jayenge par 2nd list me element bacha hua hai. uske bgle hm condition check karenge.

if (list1 == null)

currNode.next = list2

matlab jab v bacha hue element honge use next me add kar do.

ky.

if (list2 == null)

currNode.next = list1

uske baad jab bhi node create kiya tha uske next of return kar denge.